



TECHNICAL MAINTENANCE MANUAL

CCTV Maintenance Manual - UH-CCTV-604

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Client	UrbanHive Facility Technologies Pvt. Ltd.
Site	GreenPark Meridian Residency, Whitefield, Bengaluru
Effective date	2024-07-01
Review cycle	Annual or after any major incident

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1. Equipment Overview and Technical Specifications

IP surveillance system for perimeter, lobby, and parking at GreenPark Meridian. 4MP varifocal cameras, PoE switches, NVR with 90-day retention, and VLAN segmentation.

Model code: UH-CCTV-604

Asset registry: Record serial number and installation tower in AMIS before first energization.

Electrical supply: PoE+ 30W per camera, dedicated CCTV UPS 2kVA, CAT6A shielded horizontal cabling

Environmental limits: Ambient 0-45°C operational; store spare parts below 35°C and 60% RH.

Standards: UrbanHive engineering bulletin UH-ENG-AMIS-2024; applicable Indian Standards as cited per asset class.

Documentation package: This manual, wiring diagram sheet WD-UH-CCTV-604, commissioning checklist CC-041, and spare parts catalogue SP-UH-CCTV-604.

Technicians must verify nameplate data matches AMIS asset record UH-CCTV-604 before applying procedures in this document. Discrepancies must be logged as a data quality ticket to UrbanHive Data Engineering within one business day.

2. Installation Guidelines

Camera height 3.5m min for facial capture; avoid back-lighting into sun path. Record retention policy configured per UrbanHive data governance UH-DG-CCTV-01.

Pre-commissioning checks:

- Verify incoming power quality: voltage balance within 2%, THD < 5% on dedicated feeder.
- Confirm earthing resistance ≤ 1 ohm at equipment location; bond to society main earth bar.
- Inspect shipping damage; photograph defects for vendor warranty within 48 hours.
- Complete mechanical alignment, torque logs, and insulation resistance tests per UrbanHive QC form UH-QC-INST-02.

Post-installation: Upload signed commissioning certificate, as-built photos, and warranty start date to AMIS. Tag physical asset with QR code linking to this manual revision v1.0.0.

3. Preventive Maintenance Schedule

UrbanHive mandates the following intervals for GreenPark Meridian. Adjust only with written approval from Head of Maintenance.

- Daily: NVR health check, disk SMART status.
- Weekly: clean camera domes with microfiber; verify focus on entry gates.
- Monthly: firmware review; test failover to secondary NVR.
- Quarterly: cable pathway inspection for rodent damage.
- Annual: replace UPS batteries; recalibrate camera clocks to NTP.

Documentation: Each PM visit must generate a service log in AMIS with parts consumed, meter readings, and pass/fail checklist. Missed PM windows trigger automatic escalation to Facility Manager after 72 hours.

4. Troubleshooting Guide

Use isolation procedure: observe symptom ? confirm with meter/test ? apply remediation ? functional test ? document in AMIS.

If fault persists after two visits, escalate to UrbanHive vendor support with fault code and photos.

Fault 01 - Code C101

Symptom: Camera offline - no ping response

Remediation: Check PoE port power; reseal patch cord; verify VLAN 40 membership.

Fault 02 - Code C204

Symptom: Night image washout on IR

Remediation: Clean IR window; adjust exposure limit; verify IR cut filter operation.

Fault 03 - Code C310

Symptom: NVR disk degraded SMART alarm

Remediation: Replace disk bay 3; rebuild RAID5; verify write throughput > 120 MB/s.

Fault 04 - Code C405

Symptom: Recording gap detected 02:00-02:45

Remediation: Review NTP drift; inspect scheduled reboot task; check disk full threshold.

Fault 05 - Code C512

Symptom: Motion detection false triggers

Remediation: Mask tree branch zone; reduce sensitivity to level 4; enable human classification.

Fault 06 - Code C601

Symptom: Switch port flapping on uplink

Remediation: Replace SFP module; clean fiber ferrule; verify bend radius > 30mm.

Fault 07 - Code C708

Symptom: Password lockout on admin account

Remediation: Follow UrbanHive credential reset SOP; enable MFA on NVR.

Fault 08 - Code C802

Symptom: Corridor camera blurred image

Remediation: Refocus varifocal lens; clean dome interior condensation.

Fault 09 - Code C901

Symptom: Storage retention below 90-day policy

Remediation: Add 8TB disk; archive oldest week to cold storage per UH-DG-CCTV-01.

Fault 10 - Code C1003

Symptom: ONVIF stream failure to VMS

Remediation: Update ONVIF profile; verify multicast not blocked on VLAN.

Fault 11 - Code C1102

Symptom: Lightning surge damaged port 8

Remediation: Replace switch; install surge protector SP-C-SPD; earth bond check.

Fault 12 - Code C1201

Symptom: Privacy mask misaligned after PTZ move

Remediation: Recalibrate home position; redraw mask polygons in VMS.

5. Spare Parts List

Maintain minimum stock at GreenPark Meridian central store (Basement B1, Room ST-04).

Part No.	Description	Min stock
SP-C01	4MP dome camera	6
SP-C02	PoE switch 24-port	1
SP-C03	NVR 8TB disk	4
SP-C04	SFP module 1G	4
SP-C05	UPS battery 12V 9Ah	4
SP-C06	Surge protector RJ45	12

Reorder when stock falls below minimum. Use UrbanHive procurement portal PR-UH-OPS for approved vendors only.

6. Safety Warnings and Compliance

Work at height permit for pole cameras. Do not record beyond society boundary per privacy policy. Credential access logged in UrbanHive IAM. PPE: harness, hard hat.

Permit requirements: Hot work, confined space, work-at-height, and electrical isolation require digital permit in AMIS before start.

PPE minimum: Safety shoes IS 15298, helmet IS 2925, eye protection IS 5983, gloves appropriate to task.

Incident reporting: Any injury, near-miss, or property damage - complete IR-01 within 30 minutes; notify security desk ext. 9911.

7. Manufacturer Contact and UrbanHive Support

OEM technical support: +91-80-5555-0041 (08:00-20:00 IST)

Email: support-uh-cctv-604@urbanhive-vendor.example.com

UrbanHive Ops escalation: maintenance@urbanhiveops.example.com - Priority 1 response within 4 business hours

AMIS helpdesk: Log ticket category `CCTV` with asset_id from tower registry

Emergency (life safety assets): 24x7 hotline +91-80-5555-0911

Revision history: v1.0.0 - Initial release for GreenPark Meridian AMIS integration and UrbanHive contractor certification programme.

End of document MAN-041